

Myopic Sampling Can Be Maximally Suboptimal in Risk-Limiting Financial Audits

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Abstract

The one-step oracle for risk-limiting financial audits samples transactions in proportion to their monetary misstatement contributions. We prove that repeating this rule can require nearly the entire population to be reviewed when competing fully supported policies have expected review counts arbitrarily close to one. For every population size $N \geq 2$, risk limit, and finite uniform stake cap, the sharp worst-case ratio is N for the original non-control-variate betting confidence set with logical and running intersections. The construction applies to every valid predictable betting rule obeying the cap. A large transaction with a small misstatement contribution is essential for reducing uncertainty, yet the oracle postpones it. With review-cost ratio at most κ , the sharp expected-cost ratio is $1 + (N - 1)\kappa$. The supremum ranges over populations and tolerances; the results do not cover unbounded stakes or arbitrary control-variate processes.

1 One-step information and total audit effort

Shekhar et al. [1] estimate a finite population’s total monetary misstatement by adaptive sampling without replacement. Their Proposition 2 identifies a sampling rule that minimizes the variance of the next importance-weighted observation and maximizes a lower bound on one-step wealth growth. They explicitly distinguish this surrogate from the harder multistage problem of reducing the confidence-set width as quickly as possible. We quantify that distinction: even with exact contribution information, the repeated one-step rule can have the largest possible review-count ratio.

The mechanism concerns recorded value as well as misstatement. Reviewing a large transaction can remove most of the unresolved monetary mass, even when its actual misstatement is small. Contribution-proportional sampling can postpone precisely that transaction. The proof retains the probabilistic confidence set: two candidate totals survive every bounded bet until the large transaction is reviewed. The stopping time then becomes its rank in a probability-proportional sampling order.

2 Confidence sets and admissible policies

Let $\pi_i > 0$, $\sum_{i=0}^{N-1} \pi_i = 1$, be normalized recorded values and $f_i \in [0, 1]$ fixed unknown misstatement fractions. The target is $m^* = \sum_i \pi_i f_i$. Before draw t , let R_t be the remaining indices and $A_{t-1} = \sum_{s < t} \pi_{I_s} f_{I_s}$ the observed contribution. A predictable policy chooses $q_t(i) > 0$ for every $i \in R_t$, with total probability one, and observes f_{I_t} after drawing $I_t \sim q_t$. For each candidate

$m \in [0, 1]$, define

$$\begin{aligned} Z_t &= \frac{\pi_{I_t} f_{I_t}}{q_t(I_t)}, & W_0(m) &= 1, \\ W_t(m) &= W_{t-1}(m) \{1 + \lambda_t(m)[Z_t - (m - A_{t-1})]\}. \end{aligned} \quad (1)$$

Fix a betting-rule functional B choosing predictable stakes for each policy and history. The rule must make every wealth factor nonnegative under the stipulated observation model and satisfy $|\lambda_t(m)| \leq L < \infty$ uniformly over admissible instances, policies, histories, times, and candidates. Full support gives $\mathbb{E}[Z_t | \mathcal{F}_{t-1}] = m^* - A_{t-1}$, so $W_t(m^*)$ is a nonnegative martingale. The usual maximal inequality supplies simultaneous coverage at risk limit $\delta \in (0, 1)$.

Put $A_t = \sum_{s \leq t} \pi_{I_s} f_{I_s}$ and define

$$\begin{aligned} J_t &= \left[A_t, A_t + \sum_{i \in R_{t+1}} \pi_i \right], & J_0 &= [0, 1], \\ C_t &= \bigcap_{s=0}^t (J_s \cap \{m \in [0, 1] : W_s(m) < 1/\delta\}). \end{aligned} \quad (2)$$

Thus the candidates are continuous, and both logical and running intersections are part of the construction. For $0 < \varepsilon < 1$, set

$$\tau_B(q) = \min\{1 \leq t \leq N : \text{diam } C_t \leq \varepsilon\}, \quad V_B^* = \inf_{q \in \mathcal{Q}_+} \mathbb{E}[\tau_B(q)], \quad (3)$$

where $\mathcal{Q}_+$ consists of the fully supported predictable policies above. We use $\text{diam } \emptyset = 0$; an empty set can occur on a coverage-failure event. Logical exhaustion ensures $\tau_B(q) \leq N$. The same rule B is used for all policies in a comparison.

The oracle rule is

$$q_t^{\text{or}}(i) = \frac{\pi_i f_i}{\sum_{j \in R_t} \pi_j f_j}. \quad (4)$$

All worst-case comparisons below use populations with $f_i > 0$, so this rule has full support. It is a mathematical benchmark using unrevealed contributions. The competing policies in our construction need only the recorded weights. Infima in (3) avoid assigning zero sampling probability to a positive contribution. A deterministic review justified solely by the logical interval is a separate valid procedure, but is unnecessary for these comparisons.

A two-transaction example. Suppose the first stake is zero, as in the released ApproxKelly implementation [2]. Take

$$(\pi_0, \pi_1) = (3/4, 1/4), \quad (f_0, f_1) = (1/3, 1), \quad \varepsilon = 1/3.$$

The betting set after one draw is $[0, 1]$. Reviewing item 0 leaves logical width $1/4$ and stops; reviewing item 1 leaves width $3/4$ and requires the second review. Thus $\mathbb{E}_q \tau = 2 - q_1(0)$. Equal contributions make the oracle expectation $3/2$, while recorded-value-proportional sampling gives $5/4$. The fully supported choice $q_1 = (1 - \eta, \eta)$ gives $1 + \eta$, so $V_B^* = 1$. The general theorem does not require a zero first stake.

3 A sharp worst-case theorem

Theorem 1. Fix $N \geq 2$, $\delta \in (0, 1)$, a finite $L \geq 0$, and any betting rule B satisfying the preceding assumptions. Choose a positive rational ε such that

$$\varepsilon < 1/3, \quad (1 + 2L\varepsilon)^N < 1/\delta. \quad (5)$$

For any rational $\rho \in (0, 1]$, set $a = \varepsilon/2$, $w = a/(N - 1)$, and

$$\pi_0 = 1 - a, \quad f_0 = \frac{\rho w}{1 - a}; \quad \pi_i = w, \quad f_i = 1 \quad (1 \leq i < N). \quad (6)$$

These rational populations satisfy

$$V_B^* = 1, \quad \mathbb{E}[\tau_B(q^{\text{or}})] = 1 + \frac{N - 1}{1 + \rho}. \quad (7)$$

For fixed N, δ, L, B , the supremum of $\mathbb{E}[\tau_B(q^{\text{or}})]/V_B^*$ over positive-contribution populations and tolerances $0 < \varepsilon < 1$ is exactly N .

Proof. A rational ε satisfying (5) exists because the left-hand side tends to one as $\varepsilon \downarrow 0$. The population (6) is admissible: $a < 1/6$, $w \leq a$, and $0 < \rho w/(1 - a) < 1$. Its total contribution is $m^* = a + \rho w$.

Two candidates prevent early stopping. Consider $m_- = a$ and $m_+ = 5a$, separated by 2ε . Before item 0 is reviewed, a history consists of k small items, $0 \leq k < N$, and has logical interval $[kw, 1]$. Both candidates belong to it. Under (4),

$$Z_t = \sum_{i \in R_t} \pi_i f_i = m^* - A_{t-1}, \quad Z_t - (m - A_{t-1}) = m^* - m.$$

At either candidate, $|m^* - m_{\pm}| < 2\varepsilon$. Each nonnegative wealth factor is at most $1 + 2L\varepsilon$, independently of the rule's initialization or subsequent choices. At every preceding time,

$$W_t(m_{\pm}) \leq (1 + 2L\varepsilon)^t \leq (1 + 2L\varepsilon)^N < 1/\delta. \quad (8)$$

Both candidates survive every logical and betting intersection, preventing stopping. Once item 0 is reviewed, the unreviewed recorded mass is at most $a \leq \varepsilon$, which forces stopping. The stopping time is therefore exactly the rank of item 0.

Compute the rank. Successive sampling in proportion to fixed positive rates is equivalent to ordering independent exponential clocks with those rates: the first clock has the proportional distribution, and memorylessness repeats this after each selection. Here the rates are $\rho w, w, \dots, w$. Each small item precedes item 0 with probability $w/(w + \rho w) = 1/(1 + \rho)$. Writing the rank as one plus a sum of preceding-item indicators proves (7)'s expectation.

Compare with fully supported policies. Give item 0 first-draw probability $1 - \eta$ and each other item probability $\eta/(N - 1)$, then use any fully supported continuation. The first branch stops in one review by the logical interval; all other branches use at most N reviews. Hence

$$1 \leq V_B^* \leq (1 - \eta) + \eta N = 1 + \eta(N - 1).$$

Letting $\eta \downarrow 0$ gives $V_B^* = 1$. Finally, $1 \leq \tau_B(q) \leq N$ gives the universal ratio bound N , and (7) approaches it as rational $\rho \downarrow 0$. $\square$

The theorem asserts an infimum of one, without a general nonattainment claim. A fully supported policy with suitable valid bounded stakes can stop in one review on every branch; the supplement gives an exact example that even retains the true total on both branches. Nonattainment in the preceding zero-first-stake example does not extend to the full betting class.

Corollary 2 (Heterogeneous review costs). *For known costs $c_i > 0$, put $C_B(q) = \sum_{t=1}^{\tau_B(q)} c_{I_t}$ and $V_{B,c}^* = \inf_{q \in \mathcal{Q}_+} \mathbb{E}[C_B(q)]$. If $\max_i c_i / \min_i c_i \leq \kappa$, $\kappa \geq 1$, the sharp worst-case supremum of $\mathbb{E}[C_B(q^{\text{or}})] / V_{B,c}^*$ is $1 + (N-1)\kappa$. Without bounded cost heterogeneity the ratio is unbounded, already for $N = 2$.*

Proof. Assign $c_0 = 1$ and $c_i = \kappa$ for $i \geq 1$ in (6). The rank calculation and fully supported approximation give

$$\mathbb{E}[C_B(q^{\text{or}})] = 1 + \frac{(N-1)\kappa}{1+\rho}, \quad 1 \leq V_{B,c}^* \leq 1 + \eta(N-1)\kappa.$$

Thus $V_{B,c}^* = 1$, and $\rho \downarrow 0$ yields the lower bound. For any population, every audit costs at least $c_{\min} = \min_i c_i$ and at most $\sum_i c_i \leq c_{\min}[1 + (N-1)\kappa]$, proving the upper bound. For $N = 2$ and fixed $\rho > 0$, the displayed expectation diverges as $\kappa \rightarrow \infty$. $\square$

4 Interpretation and limits

The bound concerns the one-step rule of Shekhar et al., without contradicting their one-step theorem. It holds even when sampling scores equal all realized taints, since score-proportional sampling then equals (4). A small contribution need not mean small unresolved recorded value. Recorded-value-proportional sampling stops no later than item 0's rank on (6), giving

$$\mathbb{E}[\tau_B(q^{\text{propM}})] \leq 1 + \frac{a}{1-a+w}.$$

This is a fully supported, implementable comparator close to one when ε is small, without a claim of general optimality.

Confidence sequences for sampling without replacement were developed by Waudby-Smith and Ramdas [3]; the adaptive monetary-weighted construction here is from Shekhar et al. [1]. Imberg et al. [4] optimize asymptotic estimation error in active finite-population sampling, a different objective from this stopping-time ratio. Kato and Nakagawa [5] compute sequential boundaries and expected sample sizes for binary finite populations under simple random sampling; they explicitly exclude monetary-unit and variables sampling.

The finite cap is essential to (8); the argument does not cover unbounded stakes or arbitrary control-variate wealth processes. It allows the tolerance to decrease with N , L , and δ . The sharp supremum is over tolerances, not a factor- N assertion for every prescribed tolerance. Logical and running intersections are explicit assumptions.

If simultaneous intervals $\ell_i \leq f_i \leq u_i$ are supplied separately, the residual box width after reviewing a set A is $\sum_{i \notin A} \pi_i(u_i - \ell_i)$. Sorting these widths minimizes the review count for that certificate alone. This observation and its cost extension are developed in the supplement, alongside the finite-grid solver and synthetic experiments. None is needed for the sharp theorem.

Supporting material. The supplement and repository [6] contain exact rational certificates, independent checking scripts, the support correction, and implementation details. The proofs above do not depend on computation.

References

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